



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE(Advanced)
PRE-SCORE
03-12-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)

Time : 3 Hours

PAPER-1 (OPTIONAL)

Maximum Marks : 180

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their Form No. in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **17** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. A plane electromagnetic wave of wavelength λ has an intensity I . It is propagating along the positive Y-direction. The allowed expressions for the electric and magnetic fields are given by

(A) $\vec{E} = \sqrt{\frac{2I}{\epsilon_0 c}} \cos \left[\frac{2\pi}{\lambda} (y - ct) \right] \hat{k}; \vec{B} = +\frac{1}{c} \vec{E} \hat{i}$

(B) $\vec{E} = \sqrt{\frac{I}{\epsilon_0 c}} \cos \left[\frac{2\pi}{\lambda} (y - ct) \right] \hat{k}; \vec{B} = +\frac{1}{c} \vec{E} \hat{i}$

(C) $\vec{E} = \sqrt{\frac{2I}{\epsilon_0 c}} \cos \left[\frac{2\pi}{\lambda} (y - ct) \right] \hat{k}; \vec{B} = -\frac{1}{c} \vec{E} \hat{i}$

(D) $\vec{E} = \sqrt{\frac{I}{\epsilon_0 c}} \cos \left[\frac{2\pi}{\lambda} (y - ct) \right] \hat{i}; \vec{B} = \frac{1}{c} \vec{E} \hat{k}$

2. Assume that sun rises at 6 : 00 AM & sets at 6 : 00 PM. It moves such that length of shadow of a vertical object at 12 : 00 Noon is zero. An observer observes the reflected rays from surface of water $\left(\mu = \frac{4}{3} \right)$ using a polaroid. At what approximate time will he see no light at a particular orientation of polaroid?

(A) 10 : 00 AM

(B) 11 : 00 AM

(C) 9 : 30 AM

(D) 8 : 30 AM

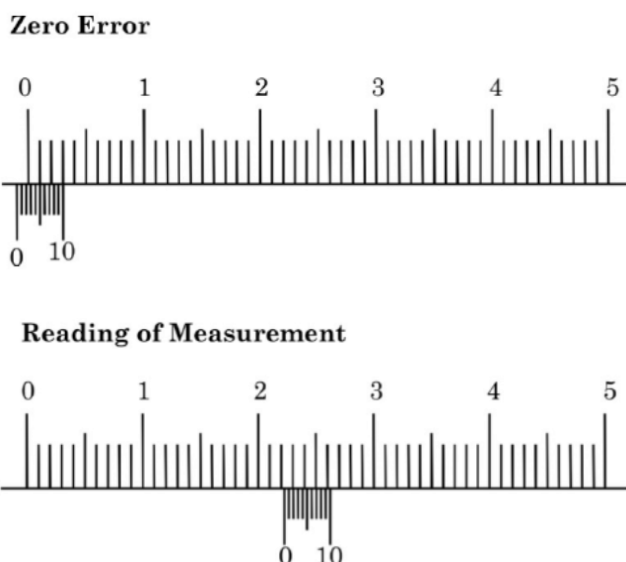
3. Longitudinal Standing wave produced in a metal rod of length 1m fixed at the left end is represented by the equation

$$y = 10^{-6} \sin \frac{\pi x}{2} \sin 200\pi t$$

where x is in metre and t is in seconds.

The maximum tensile stress at the midpoint of the rod is : (Young's modulus of material of rod = 10^{12} N/m^2)

- (A) $\frac{\pi}{\sqrt{2}} \times 10^6 \text{ N/m}^2$ (B) Zero
(C) $\frac{\pi}{2\sqrt{2}} \times 10^6 \text{ N/m}^2$ (D) $\frac{\pi}{2} \times 10^6 \text{ N/m}^2$
4. In a specially designed vernier calipers, 10 divisions of the vernier scale are equal to 4 divisions of the main scale. In the following figure, the zero error and a measurement are shown. In the zero error, the 5th and the 10th divisions of the vernier coincides with 1 mm and 3 mm mark on the main scale respectively and in the measurement, the 2nd and the 7th divisions of the vernier scale coincide with the 23 mm and 25 mm mark on the main scale respectively as shown in the figure. Report the length measured with due regards to the error.



- (A) $(21.2 \pm 0.2) \text{ mm}$
(B) $(23.2 \pm 0.2) \text{ mm}$
(C) $(23.8 \pm 0.4) \text{ mm}$
(D) $(23.4 \pm 0.6) \text{ mm}$

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

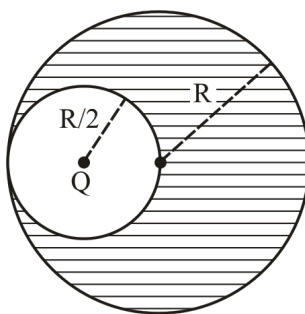
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

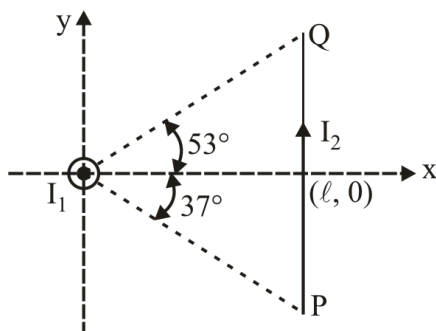
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. An uncharged conducting sphere of radius R , has a hollow cavity sphere of radius $\frac{R}{2}$, as shown. Which of the following is/are correct about electrostatic energy. A charge Q is placed at the center of the cavity.

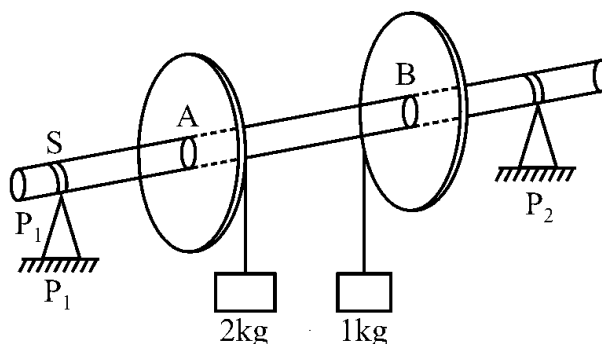


- (A) Interaction energy of induced charges with charge placed at centre of cavity is $-\left(\frac{1}{4\pi\epsilon_0}\right)\left(\frac{Q^2}{R}\right)$
- (B) Interaction energy of induced charges with charge placed at centre of cavity is $-\frac{1}{2\pi\epsilon_0}\frac{Q^2}{R}$
- (C) Electrostatic energy outside sphere is $\frac{1}{4\pi\epsilon_0}\left(\frac{Q^2}{2R}\right)$
- (D) Electrostatic energy outside sphere is $\left(\frac{1}{4\pi\epsilon_0}\right)\left(\frac{Q^2}{R}\right)$

6. An infinite wire, placed along z-axis has current I_1 in positive z-direction. A conducting rod PQ placed in xy plane parallel to y-axis has current I_2 in positive y-direction. The ends of the rod subtend angles 37° and 53° at the origin with positive x-axis as shown in figure. The rod is at a distance ℓ from the origin. Then choose the correct option(s).



- (A) The direction of net force acting on the wire PQ is along the negative z-axis.
- (B) The direction of net force acting on the wire PQ is along the positive z-axis.
- (C) The magnitude of net force acting on the rod PQ is $\frac{\mu_0 I_1 I_2}{4\pi} \ell \ln \left(\frac{16}{9} \right)$
- (D) The magnitude of net force acting on the rod PQ is $\frac{\mu_0 I_1 I_2}{2\pi} \ell \ln \left(\frac{16}{9} \right)$
7. Two discs A and B each of mass 2 kg and radius 0.5 m are rigidly fixed to the ends of a thin shaft S, which is supported in horizontal position with the help of two smooth bearings at points P_1 and P_2 . Two blocks of masses 2 kg and 1 kg are connected to the light cords wrapped around the discs as shown in the figure. There is no slipping between the cord and the discs. If the system is released from rest from the position shown, Then mark the correct statement(s). (Take : $g = 10 \text{ m/s}^2$)



- (A) Power developed by tension in cord connected to 2 kg block is negative.
- (B) Acceleration of each block is 2 m/s^2 .
- (C) Total work done by the tension on the discs is zero during the motion.
- (D) Acceleration of each block is 1 m/s^2 .

SECTION-I (iii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II**.
- **List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;

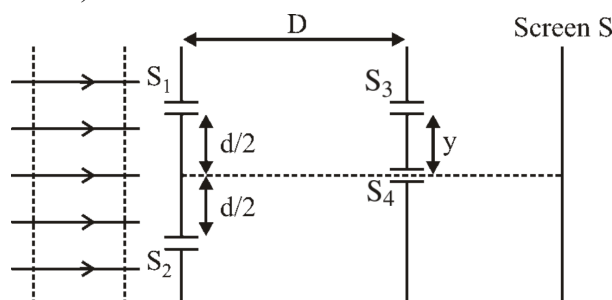
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -1 In all other cases.

8. The below diagram shows the set up for a variation of a Young's Double Slit Experiment. Plane parallel beams of two wavelengths $\lambda_1 = 2000 \text{ \AA}$ and $\lambda_2 = 7500 \text{ \AA}$ are incident on two slits S_1 and S_2 (assume S_1 and S_2 to be of identical dimension) the superposed waves from S_1 and S_2 are then incident on a second opaque screen with two slits S_3 and S_4 as shown. Superposed waves from S_3 and S_4 are then incident on a far-away screen S.

Take $\frac{D}{d} = 2 \times 10^3$, $y = 0.5 \text{ mm}$ and assume all slits to be of identical size and both beams $\lambda_1 = 2000 \text{ \AA}$ and $\lambda_2 = 7500 \text{ \AA}$ of identical intensity I_0 .

(i.e. from a single slit both S_1 and S_2 produce an intensity I_0 on S_3S_4 plane when considered one at a time).
(Also assume that S_3 and S_4 produce the same intensity on the screen S considered one at a time as they receive from $S_1 - S_2$ combination).



List-I		List-II	
(P)	Intensity of light of wavelength $\lambda_1 = 2000 \text{ \AA}$ at slit S_3	(1)	I_0
(Q)	Intensity of light of wavelength $\lambda_2 = 7500 \text{ \AA}$ at slit S_3	(2)	$2I_0$
(R)	The maximum intensity for $\lambda_1 = 2000 \text{ \AA}$ received at screen S	(3)	$9I_0$
(S)	The maximum intensity for $\lambda_2 = 7500 \text{ \AA}$ received at screen S	(4)	$\approx 11.7 I_0$
		(5)	$4I_0$

(A) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$

(B) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 3$

(C) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 3$

(D) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 4$

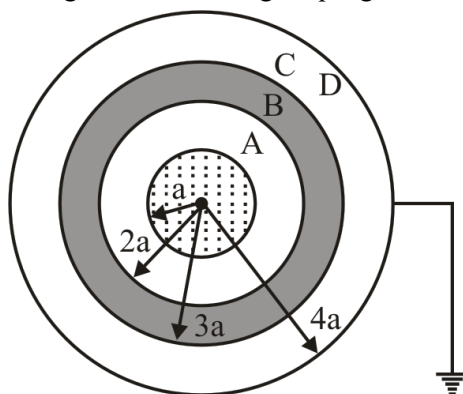
9. Two moles of an ideal monoatomic gas is taken through four different cyclic processes as depicted in the List-I and List-II gives the total work done by the gas during each cyclic process. (Take : $\gamma = 1.4$)

List-I		List-II	
(P)	In the process AB, $PT^{-\frac{1}{2}} = \text{constant}$	(1)	480R
	In the process BC, $PT^{-2} = \text{constant}$	(2)	240R
	In the process BC, $VT^{-2} = \text{constant}$	(3)	600R
	In the process CA, $VT^{-2} = \text{constant}$	(4)	540R
		(5)	300R

Which one of the following options is correct ?

- (A) $P \rightarrow 5; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 4$ (B) $P \rightarrow 5; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 2$
 (C) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 2$ (D) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 5; S \rightarrow 1$

10. A conducting sphere of radius a is placed inside a thin conducting shell of radius $4a$. Now a dielectric shell of inner radius $2a$ and outer radius $3a$ is placed between sphere and shell concentrically. A is the outer surface of the sphere, B is the inner surface of the dielectric shell, C is the outer surface of dielectric shell and D is the inner surface of conducting shell. The charge $+q$ is given to the outer surface of sphere A.



List-I		List-II	
(P)	Surface A	(1)	Charge is $-q$
(Q)	Surface B	(2)	Charge is $+q$
(R)	Surface C	(3)	Charge is $+q \left(1 - \frac{1}{k}\right)$
(S)	Surface D	(4)	Charge is $-q \left(1 - \frac{1}{k}\right)$
		(5)	None

- (A) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$
 (B) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 1$
 (C) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 2$
 (D) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 1$

11. Match the following.

List-I		List-II	
(P)	Spontaneous radioactive decay of an Uranium nucleus initially at rest as given by reaction ${}_{92}^{238}\text{U} \rightarrow {}_{90}^{234}\text{Th} + {}_2^4\text{He} + \dots$	(1)	Number of proton is increased
(Q)	Fusion reaction of two hydrogen nuclei as given by reaction ${}_1^1\text{H} + {}_1^1\text{H} \rightarrow {}_1^2\text{H} + \dots$	(2)	Momentum is conserved
(R)	Fission of U^{235} nucleus initiated by a thermal neutron as given by reaction ${}_0^1\text{n} + {}_{92}^{235}\text{U} \rightarrow {}_{46}^{144}\text{Ba} + {}_{36}^{89}\text{Kr} + 3{}_0^1\text{n} + \dots$	(3)	Mass is converted to energy or vice versa
(S)	β^- decay (negative beta decay)	(4)	Charge is conserved
		(5)	Binding energy is increased

- (A) $P \rightarrow 2,3,4,5; Q \rightarrow 2,3,4,5; R \rightarrow 2,3,4; S \rightarrow 1,2,3,4,5$
- (B) $P \rightarrow 2,3,4; Q \rightarrow 3,4,5; R \rightarrow 2,4,5; S \rightarrow 1,2,3,5$
- (C) $P \rightarrow 2,3; Q \rightarrow 2,4,5; R \rightarrow 3,5; S \rightarrow 1,2,3,4,5$
- (D) $P \rightarrow 2,3,4,5; Q \rightarrow 2,3,4,5; R \rightarrow 2,3,4,5; S \rightarrow 1,2,3,4,5$

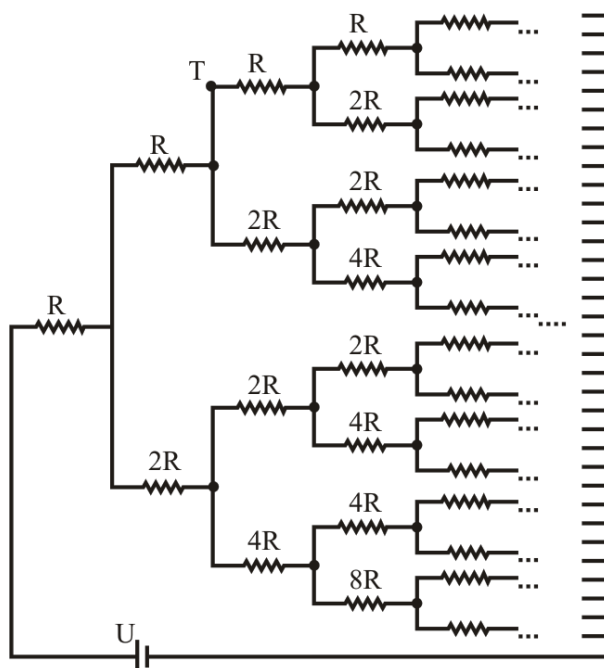
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

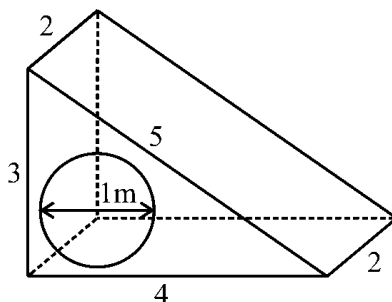
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases.

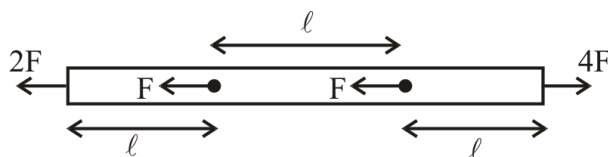
- The student assembled an "almost infinite" electrical circuit consisting of a very large number of resistors (the diagram of this circuit is shown in the figure). If you move along this circuit from left to right, then after each branching, the resistance of the resistor located above the branching point is equal to the resistance of the resistor located to the left of the branching point, and the resistance of the resistor located below the branching point is twice as large as the resistance of the resistor located above the branching point. The source voltage in this circuit is $U = 270 \text{ V}$. Find the current flowing through the cross-section of the wire at point T if $R = 20 \text{ Ohms}$.



2. A solid sphere of 1 m diameter is kept in an evacuated, enclosure as shown in the diagram. The view factor from the enclosure to the sphere is α . The value of $\left(\frac{108\alpha}{\pi}\right)$ is



3. Three kg of ice at -3°C is exposed to the atmosphere which is at 27°C . The ice melts and comes into thermal equilibrium with the atmosphere. c_p of ice is $2.0 \text{ kJ/kg } ^\circ\text{C}$ and the latent heat of fusion of ice is 333.3 kJ/kg . The change in entropy of ice (in kJ/k) is :- (Rounded off to closest integer) (Take : $\ell n(1+x) = x$ for $x \ll 1$)
4. The aphelion distance for Saturn is $1510 \times 10^6 \text{ km}$; its perihelion distance is $1350 \times 10^6 \text{ km}$. By Kepler's First Law, the Sun is at one focus of this ellipse. The distance of Sun from the other focus is $x \times 10^6 \text{ km}$. The value of x is.
5. Force acting on a uniform rod having length 3ℓ & area of cross section A and Young's modulus Y are shown in figure. find elongation in the rod (take : $\frac{F\ell}{YA} = \frac{4}{3}$)



6. A turbine operates under steady flow conditions, receiving steam at the following state : Pressure 1.2 MPa , enthalpy 3000 kJ/kg , velocity 30 m/s and elevation 3 m . The steam leaves the turbine at the following state: Pressure 20 kPa , enthalpy 2500 kJ/kg , velocity 100 m/s , and elevation 0 m . Heat is lost to the surroundings at the rate of 0.3 kJ/s . If the rate of steam flow through the turbine is 0.4 kg/s , what is the power output of the turbine in kW ? (Enthalpy is equal to total content of heat of the system equivalent to the system's internal energy plus the product of volume and pressure.) (Round off to nearest integer) ($g = 10 \text{ m/s}^2$)

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

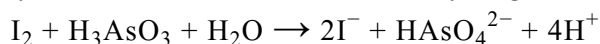
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

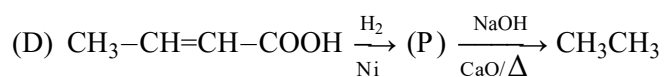
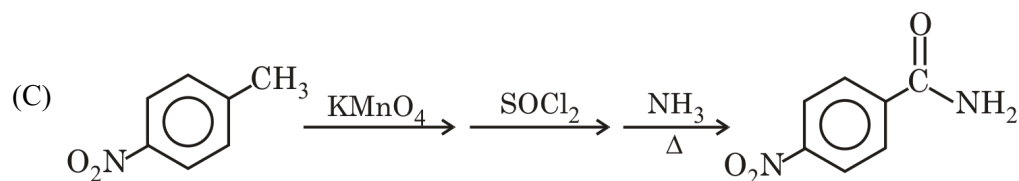
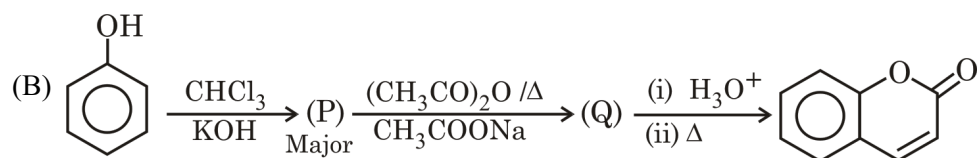
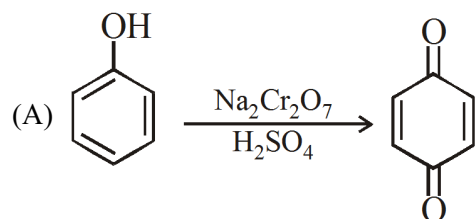
1. In an analytical determination of arsenic, a solution containing arsenious acid, H_3AsO_3 , KI and a small amount of starch is electrolysed. The electrolysis produces free I_2 from I^- ions and the I_2 immediately oxidizes the arsenious acid to hydrogen arsenate ion, HAsO_4^{2-} .



When the oxidation of arsenic is complete, the free iodine combines with the starch to give a deep blue colour. If during a particular run, it takes 96.5 s for a current of 1.68 mA to give an end point (indicated by the blue colour), how many grams of arsenic are present in the solution. [Atomic mass : As = 75]

- (A) 6.3×10^{-4} g
(B) 1.26×10^{-4} g
(C) 3.15×10^{-5} g
(D) 6.3×10^{-5} g
2. Choose the correct option
- (1) Ce(IV) can oxidise water and is a good analytical agent
(2) Ce(IV) and Yb(II) are diamagnetic in nature and are very strong oxidising agents
(3) Mischmetall contains roughly 95% lanthanoid metal and roughly 5% magnesium metal
(4) Europium has largest atomic radii among the lanthanoids
- (A) 1, 2 are correct statements
(B) 2, 3 are incorrect statements
(C) 3, 4 are correct statements
(D) 1, 4 are incorrect statements

3. Incorrect reaction sequence is :



4. In Nylon-6 which bond among following is absent :

- (A) Carbon-carbon covalent
- (B) Amide
- (C) Ester
- (D) Carbon hydrogen

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

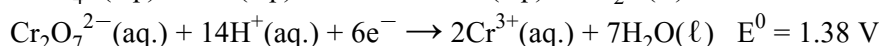
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

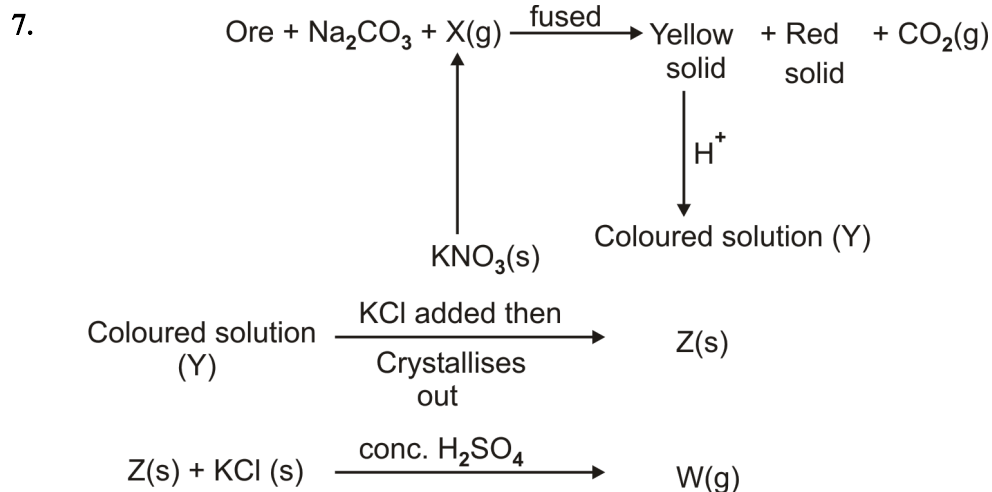
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Standard electrode potential data are useful for understanding the suitability of an oxidant in a redox titration. Some half cell reactions and their standard potentials are given below :



Identify the correct statement(s) regarding the quantitative estimation of aqueous $\text{Fe}(\text{NO}_3)_2$:

- (A) MnO_4^- can be used in aqueous HCl.
 - (B) $\text{Cr}_2\text{O}_7^{2-}$ can be used in aqueous HCl.
 - (C) MnO_4^- can be used in aqueous H_2SO_4 .
 - (D) $\text{Cr}_2\text{O}_7^{2-}$ can be used in aqueous H_2SO_4 .
6. Gold sol can be prepared by :
- (A) Peptization
 - (B) By hydrolysis
 - (C) Bredic's arc method
 - (D) Reduction



- (A) Ore used contains two metals and is magnetic in nature
- (B) The same gas (X) is evolved when $\text{KClO}_3(\text{s})$ is heated
- (C) Coloured solution (Y) on treatment with hydrogen peroxide in acidic medium, produces a blue colour compound
- (D) Solution obtained by reaction of $\text{W}_{(\text{g})} + \text{NaOH}_{(\text{aq.})}$ gives green coloration with acetaldehyde in acidic solution

SECTION-I (iii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II**.
- **List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -1 In all other cases.

8. For the reaction

$2A(g) \rightarrow B(g)$, starting with 4 moles of A in a 5 litre container at 400 K.

For all different possible orders of reaction the initial half-life is 40 seconds. Now match following :

	List-I		List-II
(P)	If reaction follows first order kinetics then initial rate of reaction	(1)	$3.46 \times 10^{-4} \text{ M/s}$
(Q)	If reaction follows zero order kinetics then rate of reaction at time : $t = 40 \text{ sec.}$	(2)	$6.93 \times 10^{-3} \text{ M/s}$
(R)	If reaction follows first order kinetics then rate of reaction at 95% completion.	(3)	$5 \times 10^{-3} \text{ M/s}$
(S)	If reaction follows second order kinetics then rate of reaction at 99% completion	(4)	10^{-6} M/s
		(5)	$6.25 \times 10^{-5} \text{ M/s}$

(A) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 1; S \rightarrow 4$

(B) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

(C) $P \rightarrow 5; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$

(D) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$

9. Choose the correct option

	List-I (Reactions)		List-II (Observations)
(P)	Ferric chloride + Potassium sulphocyanide on reaction produces the following observation	(1)	$[\text{Fe}(\text{SCN})]^{2+}$
(Q)	$\text{CoS} + \text{aqua regia}$, after this, solution is treated with potassium nitrite after neutralisation with NH_4OH and the solution is acidified with CH_3COOH produces the following observations in the process	(2)	NO gas
(R)	Lead sulphate on reaction with ammonium acetate forms a soluble complex and this complex on reaction with potassium iodide produces the following observation	(3)	Yellow ppt.
(S)	Deep red fumes turns colourless KOH solution to yellow colour solution, On treating this solution with dil. sulphuric acid and $\text{H}_2\text{O}_2 + \text{pyridine}$ produces the following observation	(4)	Brown ppt.
		(5)	Blue ppt.

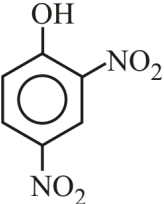
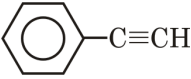
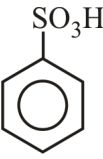
(A) $\text{P} \rightarrow 1,5; \text{Q} \rightarrow 2,5; \text{R} \rightarrow 3; \text{S} \rightarrow 1$

(B) $\text{P} \rightarrow 2,5; \text{Q} \rightarrow 3,4; \text{R} \rightarrow 5; \text{S} \rightarrow 3$

(C) $\text{P} \rightarrow 1; \text{Q} \rightarrow 2,3; \text{R} \rightarrow 3; \text{S} \rightarrow 5$

(D) $\text{P} \rightarrow 1; \text{Q} \rightarrow 3,5; \text{R} \rightarrow 4; \text{S} \rightarrow 5$

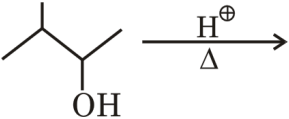
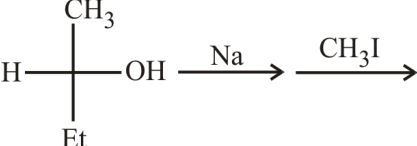
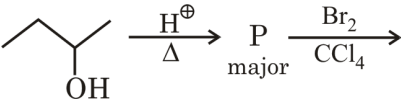
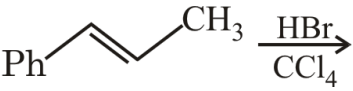
10.

List-I (Compound)		List-II (Statement)	
(P)		(1)	Is more acidic than carbonic acid
(Q)		(2)	Will release CO ₂ gas with aq. NaHCO ₃
(R)		(3)	Is soluble in aq. NaOH
(S)		(4)	Is more acidic than picric acid
		(5)	Will release H ₂ gas with Na metal

The correct code is :

- (A) P → 2,3; Q → 4,5; R → 1,4; S → 2,3,5
- (B) P → 3,4; Q → 2,3,4,5; R → 1,2; S → 2,3
- (C) P → 1,2,3,5; Q → 3,5; R → 5; S → 1,2,3,4,5
- (D) P → 3,4,5; Q → 1,2,5; R → 2,3,4; S → 1,3,5

11.

List-I (For the given reaction)		List-II (For final product or product mixture)	
(P)		(1)	Optically active product mixture
(Q)		(2)	Optically inactive product / product mixture
(R)		(3)	Product can exhibit optical isomerism
(S)		(4)	Carbocation is involved in the reaction sequence
		(5)	Racemic mixture is obtained

(A) P $\rightarrow$ 2,4; Q $\rightarrow$ 1,3; R $\rightarrow$ 2,3,4; S $\rightarrow$ 2,3,4,5

(B) P $\rightarrow$ 1,2,3; Q $\rightarrow$ 1,3,4; R $\rightarrow$ 1,2; S $\rightarrow$ 4,5

(C) P $\rightarrow$ 4,5; Q $\rightarrow$ 1,2,3; R $\rightarrow$ 3,4; S $\rightarrow$ 2,4

(D) P $\rightarrow$ 1,2,3; Q $\rightarrow$ 1,4; R $\rightarrow$ 2,5; S $\rightarrow$ 1,2,3

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

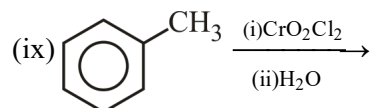
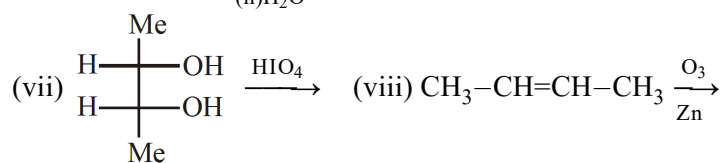
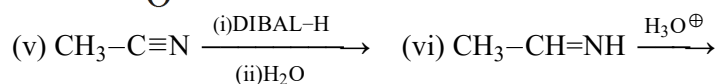
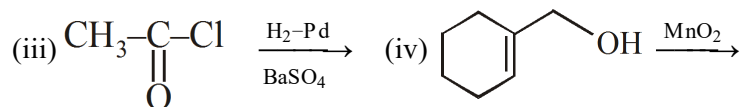
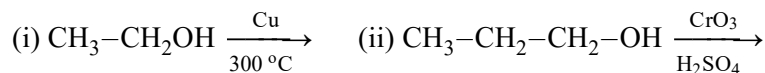
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases.

1. Silver ion forms a complex ion with thiosulphate ion. $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$, with $K_f = 2.8 \times 10^{13}$. How many moles of AgBr ($K_{sp} = 5.4 \times 10^{-13}$), will dissolve in 100 litre of 0.20 M $\text{Na}_2\text{S}_2\text{O}_3$ solution. ($\sqrt{15.12} = 3.88$) (Round off your answers to nearest integer)
2. You need to convert 1.0 mole of X into product by reaction that follows first order kinetics. At 1.00 p.m. you start the reaction at 27°C with one mole of X. At 2.0 p.m. you find that 0.8 mole of X remains. You immediately increase the temperature to 37°C . At 3.0 p.m. you discover that 0.4 mole of X are still remaining. You want to be able to finish the reaction by 4.0 p.m. but you cannot stop the reaction until only 0.1 mole of X remains. You decided to increase the temperature again. What is the minimum temperature (in K) required to complete the reaction by 4.0 p.m. ? [Use : $\ln 2 = 0.7$, $\ln 5 = 1.6$, $\ln 7 = 1.9$, $R = 8\text{J/K-mole}$] (Round off your answer to nearest integer)
3. Number of correct statement(s) :-
 - (1) Zn , Cd , Hg and Mn have typical metallic structure at normal temperature
 - (2) Zn has the lowest enthalpy of atomization among the 3d series metals
 - (3) In general density of 3d series transition metals increases as we move more from left to right in a period
 - (4) Cr^{+2} is reducing and Mn^{+3} is oxidizing in nature
 - (5) Cu^{+1} compounds in aqueous solution undergoes disproportionation reaction
 - (6) Ferrates readily decomposed into ferric oxide and $\text{O}_2(\text{g})$
 - (7) In general tendency to form divalent cations across the 3d series decreases because the sum of the $\text{IE}_1 + \text{IE}_2$ increases as we move from left to right in a period
 - (8) Interstitial compounds are chemically inert and they retain their metallic conductivity
4. 'X' are the number of complexes which are stable than $[\text{Co}(\text{CN})_6]^{3-}$ among the following $[\text{CoCl}_6]^{4-}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Co}(\text{ox})_3]^{3-}$, $[\text{Co}(\text{en})_3]^{3+}$
 'Y' are number of linkage isomers of $[\text{Co}(\text{SCN})_4(\text{NO}_2)_2]^{3-}$
 'Z' are number of compounds which show geometrical isomerism among the following compounds $[\text{Pt}(\text{en})_2]^{2+}$, $[\text{Pt}(\text{en})\text{Cl}_2]^{2+}$, $[\text{Pt}(\text{gly})\text{Cl}_2]$, $[\text{Pt}(\text{bn})_2]^{2+}$
 Find $\frac{X+Y}{Z}$.

5. (i) Boric acid on heating yields a compound Y, Y on heating yields a compound Z, the oxidation state of Z is 'A'.
 (ii) Metal Borohydrides are prepared by the reaction of metal hydrides with compound 'C' in diethyl ether, number of hybrid orbitals involves in one molecule of product formed is 'B'.
 Find A + B

6. Number of reactions in which aldehyde is obtained by oxidation of organic reactant ?



PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

-
1. The graph of $2x^2 + xy + 3y^2 - 11x - 20y + 40 = 0$ is an ellipse in the first quadrant of the xy -plane. Let a and b be the maximum and minimum values of $\frac{y}{x}$ over all points (x, y) on the ellipse. What is the value of $a + b$?
- (A) 3
- (B) $\sqrt{10}$
- (C) $\frac{7}{2}$
- (D) $\frac{9}{2}$
2. How many 15-letter arrangements of 5 A's, 5 B's and 5 C's have no A's in the first 5 letters, no B's in the next 5 letters and no C's in the last 5 letters ?
- (A) $\sum_{k=0}^5 \binom{5}{k}^3$
- (B) $3^5 \cdot 2^5$
- (C) 2^{15}
- (D) $\frac{15!}{(5!)^3}$

3. If $f(x) = 3 \left\{ \frac{[x]}{3} \right\}$, $g(x) = \left[\frac{1}{1+x^2} \right]$ and $\phi(x) = g(f(x))\{x\}^2 + g(f(x) - 1) + g(f(x) - 2)(1 - \{x\})$

where $[x]$ and $\{x\}$ denote integral and fractional parts of x respectively, then $\int_0^{2024} \phi(x) dx =$

- (A) 1235
(B) 1233
(C) $1227\frac{1}{3}$
(D) $1228\frac{1}{3}$
4. The image of the point $(3, 5)$ in the line $x - y + 1 = 0$, lies on :
- (A) $(x - 2)^2 + (y - 2)^2 = 12$
(B) $(x - 2)^2 + (y - 4)^2 = 4$
(C) $(x - 4)^2 + (y + 2)^2 = 16$
(D) $(x - 4)^2 + (y - 4)^2 = 8$

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Which of the following options are correct ?

(A) If $a \geq b > 1$, then the largest possible value of $\log_a \left(\frac{a}{b} \right) + \log_a \left(\frac{b}{a} \right)$ is 0

(B) Number of integers 'b' for which $\log_b 2024$ is a positive integer is 0.

(C) There is a unique positive integer 'n' such that $\log_2(\log_{16} n) = \log_4(\log_4 n)$. The sum of the digits of 'n' is 13.

(D) The set of real numbers 'x' for which $\log_{2024} \log_{2023} \log_{2022} \log_{2021} \log_{2020} x$ is defined is $\{x | x > c\}$. Then 'c' is $2020^{2021^{2022}}$

6. Let $f(x)$ and $g(x)$ are differentiable functions such that $f(x) + \int_0^x g(t) dt = \sin x(\cos x - \sin x)$, and $(f(x))^2 + (g(x))^2 = 1$ then $f(x)$ and $g(x)$ respectively, are
- (A) $\cos^2 x, \cos 2x$
- (B) $\frac{\cos 2x}{2}, \cos 2x$
- (C) $\frac{1}{2} \sin 2x, -\sin 2x$
- (D) $-\sin^2 x, \cos 2x$
7. If $f(x)$ be such that $f(x) = \max(|3 - x|, 3 - x^3)$ then :
- (A) $f(x)$ is continuous $\forall x \in \mathbb{R}$
- (B) $f(x)$ is derivable at three points only
- (C) $f(x)$ is non-derivable at three points only
- (D) $f(x)$ is non-derivable at four points only

SECTION-I (iii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II**.
- **List-I** has **Four** entries (P), (Q), (R) and (S) and **List-II** has **Five** entries (1), (2), (3), (4) and (5).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -1 In all other cases.

8. Match the following from List-I to List-II.

List-I		List-II	
(P)	The equation of tangent of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ which cuts off numerically equal intercepts on axes is $x - y = a$ where $ a $ equal to	(1)	$\sqrt{2}$
(Q)	The normal $y = mx - 2am - am^3$ to the parabola $y^2 = 4ax$ subtends a right angle at the vertex if $ m $ equal to	(2)	$\sqrt{3}$
(R)	The equation of the common tangent to parabola $y^2 = 4ax$ and $x^2 = 4y$ is $x + y + \frac{n}{\sqrt{3}} = 0$, then n is equal to	(3)	$\sqrt{8}$
(S)	If the equation of common tangent to parabola $y^2 = 8x$ and the hyperbola $3x^2 - y^2 = 3$ is $4x - 2y + \frac{\lambda}{\sqrt{2}} = 0$, then $ \lambda $ is equal to	(4)	$\sqrt{41}$
		(5)	2

(A) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

(B) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 1; S \rightarrow 3$

(C) $P \rightarrow 2; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 3$

(D) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$

9. Match the following.

List-I		List-II	
(P)	$(2x - 10y^3) \frac{dy}{dx} + y = 0$, $y(1) = 1$ then the value of $[x]$ when $y = 2$ is equal to ([.] GIF)	(1)	3
(Q)	$(x^2y^2 - 1)dy + 2xy^3dx = 0$, $y(1) = 1$ then at $y = 2$ the value $4x^2$ equals	(2)	7
(R)	Area of the region formed by the points (x, y) satisfying $[x]^2 = [y]^2$ for $0 \leq x \leq 4$ is greater than ([.] GIF)	(3)	15
(S)	Let the function $f(x)$ which is continuous every where, and satisfies $f(x) + \frac{1}{f(x)} = e^x + e^{-x}$ then the area bounded by $y = f(x)$, x -axis, $x = -1$ and $x = 1$ can have integral part	(4)	2
		(5)	1

(A) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 1, 4, 5; S \rightarrow 4$

(B) $P \rightarrow 2; Q \rightarrow 3; R \rightarrow 2, 3, 4; S \rightarrow 5$

(C) $P \rightarrow 5; Q \rightarrow 1, 2, 3; R \rightarrow 4; S \rightarrow 2$

(D) $P \rightarrow 3, 4, 5; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 3$

10. Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(P)	Number of real roots of $2^{-x} + 2^{-x^3} - x = 0$ is	(1)	2
(Q)	Let $\theta_1 + \theta_2 = 2\theta$ (where θ is acute angle) then $\frac{2(\sin \theta_1 - \sin \theta_2)}{(\cos \theta_2 - \cos \theta_1)} \tan \theta$ is (wherever defined)	(2)	3
(R)	$6 \left(\log_{20} \left(\sqrt{6 - 2\sqrt{5}} + \sqrt{6 + 2\sqrt{5}} \right) \right)$ is	(3)	4
(S)	Number of integers satisfying the inequality $\log_{\frac{1}{e}} x - 4 > -1$ is	(4)	1
		(5)	6

The correct option is :

(A) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 3$

(B) $P \rightarrow 1; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 4$

(C) $P \rightarrow 4; Q \rightarrow 2; R \rightarrow 3; S \rightarrow 1$

(D) $P \rightarrow 1; Q \rightarrow 4; R \rightarrow 3; S \rightarrow 2$

11. Match the following

List-I		List-II	
(P)	The value of $\int_0^1 \sqrt[5]{2x^3 - 3x^2 + 3x - 1} dx$ is less than	(1)	0
(Q)	Let $I_1 = \int_0^{\pi/2} \cos\left(\frac{2\pi}{3}\sin^2 x\right) dx$, $I_2 = \int_0^{\pi/2} \cos\left(\frac{\pi}{3}\sin x\right) dx$ then the value of $\frac{I_2}{I_1}$ is greater than	(2)	1
(R)	$\int x(x+1)\left(\frac{1}{x^8} + \frac{2}{x^9}\right)^{1/10} dx = \phi(x)$, $\phi(0) = 0$, $\phi(1) = \frac{a}{11}(3)^{\frac{11}{b}}$ then $\frac{b}{a}$ is less than	(3)	2
(S)	Let $f(x) = \int x^2(2x + 6\tan x - 2x\tan^2 x)\cos^2 x dx$ and $y = f(x)$ passes through $(\pi, 0)$ then the number of solutions of $f(x) = x^3$ is less than	(4)	3
		(5)	4

- (A) $P \rightarrow 1; Q \rightarrow 2, 3; R \rightarrow 3, 4, 5; S \rightarrow 1, 2$
 (B) $P \rightarrow 2, 3, 4, 5; Q \rightarrow 1, 2; R \rightarrow 4, 5; S \rightarrow 5$
 (C) $P \rightarrow 3; Q \rightarrow 2, 4, 5; R \rightarrow 3, 4; S \rightarrow 1, 2$
 (D) $P \rightarrow 1, 2; Q \rightarrow 1, 3, 5; R \rightarrow 4; S \rightarrow 1, 5$

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases.

- Two distinct, real, infinite geometric series each have a sum of 1 and have the same second term. The third term of one of the series is $1/8$, and the second term of both series can be written in the form $\frac{\sqrt{m}-n}{p}$ where m , n and p are positive integers and m is not divisible by the square of any prime. Find $100m + 10n + p$.
- Let $P(x)$ be a quadratic polynomial with real coefficients satisfying $x^2 - 2x + 2 \leq P(x) \leq 2x^2 - 4x + 3$ for all real numbers x , and suppose $P(11) = 181$. Find $P(16)$.
- Find $L = \lim_{x \rightarrow \infty} \left(\sqrt{x} \sin \left(\sin \left(\sin \left(\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{\sqrt{xn^2 + k}} \right) \right) \right) \right)$
- Let $A_n = [a_{ij}]$ be a square matrix of order 3, where $a_{ij} = \frac{2i+j}{3^{2n}}$ for all i and j , $1 \leq i, j \leq 3$. If $\ell = \lim_{n \rightarrow \infty} \text{Tr.} (3A_1 + 3^2A_2 + 3^3A_3 + \dots + 3^nA_n)$, then ' ℓ ' is equal to (where $\text{Tr.}A$ means trace of A)
- If the ratio of the 4th term to the 2nd term in the expansion of $\left(\frac{1}{x} + x \tan x \right)^5$ is $\frac{2}{27} \pi^4$ then the value of $[x]$ _____
([.] denotes G.I.F.) where $x > 0$
- Let $f(x) = x^\alpha + 3x^\beta$ with $\alpha > 1$, $\beta > 1$. If the value of a for which the area of the region bounded by $y = f(x)$ and straight lines $x = 0$, $x = 1$ and $y = f(a)$ is minimum is λ , then the value of 70λ is

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NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator**Space for Rough Work****ALLEN CAREER INSTITUTE Pvt. Ltd.**

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